Mario Casas Donjuan

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PROFESSIONAL SUMMARY

Robotics and Digital Systems Engineering student working across embedded electronics, control systems, and software automation.Leads an avionics team developing electronics and telemetry for aerospace applications, and mentors programming on a robotics team, previously building Java control systems for competitive robotics (three World Championship qualifications). Delivered a workflow automation system for an internal SAP team in six weeks. Experience in PCB design (Altium Designer), Python services with computer vision, and self-hosted infrastructure.

EDUCATION & CERTIFICATIONS

Tecnológico de Monterrey Monterrey, NL
B.S. in Robotics and Digital Systems Engineering *Aug. 2025 – Jun. 2029 (Exp.)*
Grade Average: 88/100 (Preliminary)

Technical Certifications Online
Altium Education Course Quiz *2026*

Prepa Tec Eugenio Garza Lagüera Monterrey, NL
Multicultural Baccalaureate (IB) | Average: 88/100, Honors Program, Global Citizenship Stole *Aug. 2022 – Jun. 2025*

TECHNICAL EXPERIENCE

Monterrey Rocket Lab (MRL) Monterrey, NL
Avionics Engineer & Team Leader *Sep. 2025 – Present*
Lead the avionics team, developing and integrating electronic systems and control logic for aerospace applications.

Build telemetry systems for real-time data acquisition and transmission during flight.

SAP – Talent Lab Program Monterrey, NL
Software Intern, Workflow Automation *Jun. 2026 – Jul. 2026*
Automated a manual international call scheduling workflow using Power Automate, SharePoint, and Microsoft Teams, including UTC-overlap time zone matching and Adaptive Card notifications with dynamic @mentions.

Gathered requirements from stakeholders and authored technical documentation enabling the team to maintain and extend the system, under a 6-week delivery constraint.

FIRST Robotics Competition – Team Voltec 6647 Monterrey, NL
Programming Mentor / Former Team Leader & Main Driver *Aug. 2022 – Present*
Developed Java control systems with PID algorithms and computer vision for 60 kg robots; Multiple Regional Winner and 3x World Championship qualifier.

Mentor junior developers on software architecture and OOP best practices.

PROJECTS

PCB Redesign – Monterrey Rocket Lab | ***Altium Designer, Schematic Capture*** 2026
– Redesigned a Monterrey Rocket Lab PCB (originally in EasyEDA) in Altium Designer, reworking schematic and layout.

MichiBeam Automated System | ***ESP32, C++, 3D Printing, CAD*** Feb. 2026 – Mar. 2026
– Engineered an autonomous hardware-software system with microcontrollers and servomotors, designing physical components in CAD and applying mathematical modeling for control logic.

Industrial IoT Startup Development | ***Systems Architecture, B2B*** Apr. 2026 – Present
– Designed the software infrastructure and data transmission pipeline for remote cold-chain monitoring.

RepLab – Exercise Form Analysis | ***Python, FastAPI, OpenCV, MediaPipe, SQLite*** May 2026
– Built a computer-vision pipeline that segments repetitions and computes a 0–100 form score, served via FastAPI with Gemini API feedback. Built in a one-day hackathon.

Home Lab Server Architecture | ***Proxmox VE, Linux, Docker, Networking*** Jan. 2025 – Present
– Deployed and maintain a bare-metal Proxmox VE server hosting services in Docker containers, with secure network tunnels (VLANs, Tailscale) for remote access.

SKILLS & LANGUAGES

Programming: Python (Advanced), Java (Advanced), C++ (Advanced), SQL (Intermediate), TypeScript, MATLAB, C (Beginner).
Hardware & Embedded: Altium Designer (Basic), PCB design and schematic capture, ESP32 microcontrollers, PID control, Onshape (CAD).
Data & Automation: Power Automate, SharePoint, Excel, Pandas, scikit-learn, FastAPI (Intermediate).
Infrastructure & Tools: Linux, Docker, Git/GitHub (Intermediate), Proxmox VE, Tailscale, VS Code, Neovim, LaTeX.
Languages: Spanish (Native), English (C1 – Advanced), German (A1), French (A2).